Lecture 3

Environmental stress and yield

Plant population

Total number of plants in a given area

Square metres

Limas

Hectares

Done by actual counting or estimate

Acres

Plant population

Maize

0.3m X 0.75m = 0.225m2

1 hectare - 100m X 100m = 10,000m2

10,000m2/0.225m2 = 44,444.44 plants

1 lima – 50m X 50m = 2,500m2

2,500m2/0.225m2 = 11,111.11 plants

Cabbage

0.45m X 0.9m = 0.405m2

1 hectare – 10,000m2

10,000m2/ 0.405m2 = 24,692 plants

1 lima – 2,500m2

2,500m2/ 0.405m2 = 6,173 plants

Soil moisture

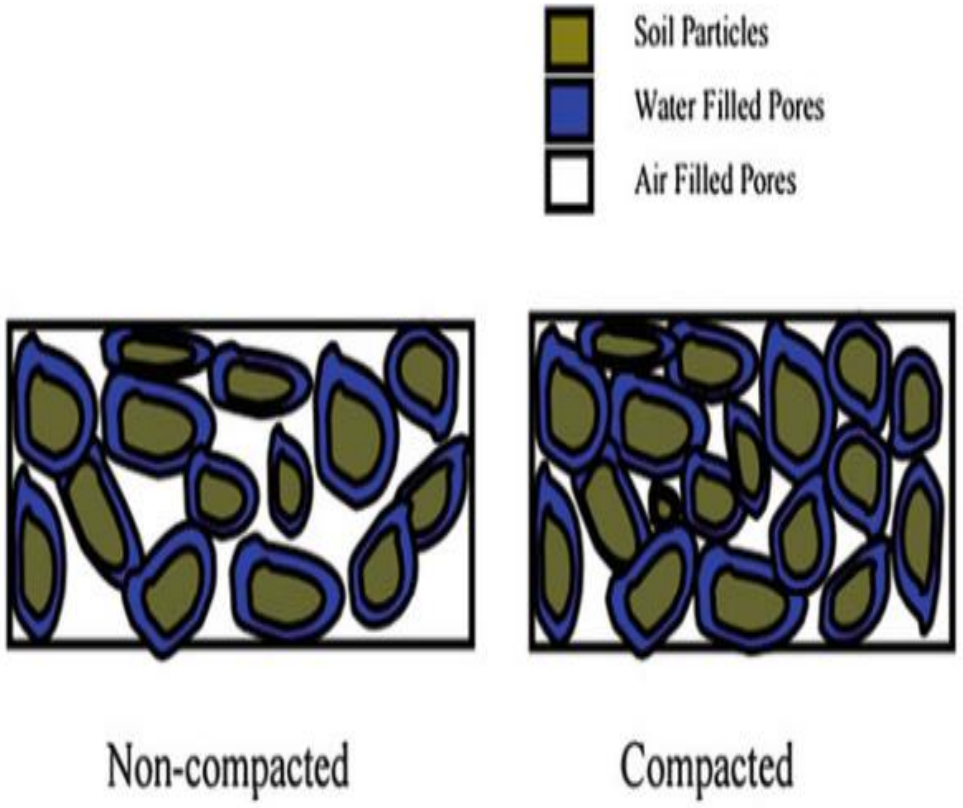
This is the quantity of water in the soil. Depends on the pore space in the soil.

The reduction of soil volume (a simple reduction in pore space) due to external factors is called soil compaction

Wet soils

Field capacity

Dry soil



Crop mixtures

Simultaneous growing of two or more crops mixed together and then sown either in lines or they are broadcasted is called crop mixture

This type of intercropping system is used for grass–legume

Benefit

Maximise on the area

Problem

Inhibit operations in the field

Relay cropping

Simultaneous growing of two or more crops during part of the life cycle of each is called relay intercropping

When the first crop has reached its reproductive stage but before it is ready for harvest, a second crop is planted so that the first crop is harvested to make room for the full development of the second

Crop rotation interchange of crops grown in different seasons

Agroforestry mixture of plant trees with for fertility

weeds

Plants growing where they are not wanted

Plants can be classified as weeds if they cause economic loss to a valuable crop, are harmful to humans and livestock, or are detrimental to wildlife and ecosystems

Invasive plants are nonnative plants that become established in natural plant areas and replace native vegetation

Most weeds affecting crop production are terrestrial and live on land

Other weeds such as water hyacinth are aquatic

Weed classification

Like crop plants, weeds are annual, biennial, or perennial

Monocot weeds affecting crop production systems are members of the grass and sedge families

Dicots are frequently called broadleaves plants that affect production of crops

Weed competition

Weeds compete with crops for light, water, and minerals

Weeds increase the costs of crop production. The traditional approach to weed control using tillage required considerable labor inputs

Mixing weed seed with grain seed can result in economic loss if separation is not possible at harvest or later during processing

Weeds can interfere with harvesting operations such as combine harvester

Weed survival dormancy, dispersal, vegetative growth,

Soil fertility

Soil organic matter (SOM) benefits the soil and the health of plants

Soil organic matter (SOM) is any material that is produced originally by living organisms (plant or animal) which is returned to the soil that undergoes the decomposition process and releases plant nutrients

The soil organic matter contains three main components, namely, biomass (living organisms), detritus (dead tissues and waste), and humus (nonliving and non-tissue)

Soil fertility is built and maintained by retaining crop residues, growing cover/green manure crops, crop rotation, and addition of compost and farmyard manure

Fertility improvement

Manures and fertilizers are applied to the soil to improve fertility contains one or more of the essential elements, e.g., nitrogen, phosphorus, potassium, calcium, magnesium

manure	Fertilizer
Organic material	Chemical substance
Natural material	Inorganic substance
Humus addition	No humus addition
Infield preparation	Manufacture in factory

Pests

Insects

Types of insects

Aphids,

caterpillar,

spider mites, thrips, flies, bettles

Nematodes

Plant diseases

Fungi are organisms composed of filamentous threads called hyphae that absorb nutrients. The hyphae grow and form a mycelium, a web-like mass that makes up the body of the fungus

Typical fungal disease symptoms include leaf spots, blights, wilts, rots, and cankers

A group of fungi called basidiomycetes cause rust diseases

Powdery mildew is a very common disease found in a wide range of plants

Bacteria are microscopic, single-celled organisms. Most plant pathogenic bacteria are rod-shaped

Crown gall is a widespread and economically important bacterial disease

Bacterial wilt

Viruses are extremely small, simple pathogens that are composed of nucleic acid

Tobacco mosaic virus

Maize mosaic virus